



Jargon Buster

MPEG: MPEG is a group (Moving Picture Experts Group) that develops standards for digital audio and video compression. There are several versions of this standard, including MPEG-1, MPEG-2 and MPEG-4.

MPEG-1: This standard was designed to code progressive video at a transmission rate of 1.5 Mbps. It is used in Video CDs, but is giving way to the new MPEG-2 standard. MPEG-1 Layer 3 is the audio compression format popularly known as MP3.

MPEG-2: This standard was designed to code transmission rates above 4 Mbps, and is used in DVD and digital broadcast TV. It is superior to MPEG-1 in terms of picture quality and sound clarity.

MPEG-4: This is a relatively new standard, and is aimed at converging streaming media from different sources. Speech and video synthesis, fractal geometry and an AI approach to reconstruct images with better, yet lossless, compression.

NTSC: This stands for National Television Standards Committee that developed the protocol for broadcast transmission and reception in the US. NTSC signals are used in the US and Japan, and have hardly been altered since their inception, except for the addition of new parameters for colour signals. NTSC signals are

interlaced, and an NTSC TV image has 525 horizontal lines per frame. Every other line is dropped, and thus it takes two screen scans to complete one image. One complete frame is scanned every 1/30 second. NTSC signals are not directly compatible with computer systems, but there are adapters available that let you view an NTSC signal on a PC monitor.

PAL: Phase Alternation Line is the standard that is used in India and Europe. In the PAL standard, the horizontal image has 625 horizontal lines per frame. A slight colour variation is seen between the PAL and NTSC standards. This standard was developed in Germany.

SECAM: This stands for Sequential Couleur avec Memoire, and is prevalent in some parts of Europe—mainly in Russia and countries of the former Soviet Union. Similar to the PAL standard, it is an interlaced transmission signal for TV.

NICAM stereo: This stands for Near Instantaneous Companded Audio Multiplex, and was developed by the BBC Research Centre in the early 1980s. It was first transmitted with the PAL colour broadcasting system in Britain. This technology improves on the sound quality of the transmitted TV signal. The Compro and Pinnacle TV tuners feature NICAM stereo.



CATEGORY		TV TUNER CARD				
Brand	Aviosys	Compro	Mercury	Pinnacle	Pinnacle	PixelView
Model	TV-EZ	TV Gold+	TV Tuner	PC TV Plus	PC TV PRO	PlayTV MPEG-2
Chipset	Philips 7134	Philips 7134	Philips 7130	Conexant BT878	Conexant BT878	Conexant BT878A
Internal / External	Internal	Internal	Internal	Internal	Internal	Internal
Video Capture formats supported (MPEG1 / MPEG2)	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
Additional Video Capture Device Connector	✗	✗	✗	✗	✗	✗
Colour Formats supported (YUV / YUV12 / RGB16 / 32)	✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓	✓ / ✓ / ✓ / ✓
Maximum resolution supported	720 x 576	720 x 576	720 x 576	720 x 576	720 x 576	720 x 576
FM Tuner	✓	✓	✓	✗	✓	✓
Time-shifting software	✓ / ✓	✓	✓	✓	✓	✓
TV Standards Supported (NTSC / PAL / SECAM)	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓
Other features	✗	NICAM Stereo	✗	NICAM Stereo	NICAM Stereo	✗
Package Contents (15%)						
Software bundled	VCD DVD Plus, Ulead VideoStudio	Ulead DVD Movie Factory, Ulead VideoStudio, Compro Internal software	Mercury proprietary software	Pinnacle proprietary PCTV Vision, Studio 7 Trial, T-Rex Multimedia file converter	Pinnacle proprietary PCTV Vision, Studio 7 Trial, T-Rex Multimedia file converter	Cyberlink PowerVCR
Manuals / Remote Control	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
Ease Of Use (So5 *) (15%)						
Installation / Driver support / Configuration	4 / 4 / 4	4 / 4 / 4	3 / 3 / 3	4 / 4 / 4	4 / 4 / 4	3 / 3 / 3
Quality Of Encoding (Dropped Frames)	0	0	3	0	0	0
Number of channels detected	78	78	60	78	78	78
Fine tuning required	✗	✓	✓	✗	✗	✗
Processor Usage (%)	21	21	22	21	21	21
Clarity of captured content / Overall Quality (So5 *)	4 / 4	4 / 4.5	2.5 / 2	4 / 4	4 / 4	3 / 3
Performance	19.70	19.50	11.02	19.70	20.50	16.50
Features	58.50	62.50	46.00	55.50	61.00	53.50
Grand Total (Performance + Features)	78.20	82.00	57.02	75.20	81.50	70.00
Value for money	3.13	2.09	3.35	2.51	2.33	2.00
Price (Rs)	2,500	3,925	1,700	3,000	3,500	3,500
Grade	B+	B-	B-	B	B-	B-

* Scale of 5

